Mini-GPT: Design

The architectural decisions behind Mini-GPT, with the reason for each.

16 October 2026

Part	Decision
Data	FineWeb-Edu sample-10BT, documents shuffled globally into one token stream
Validation	1,912 held-out documents (8,569,151 UTF-8 bytes), chosen by a hash of the doc id, kept as raw text
Tokenizer	Byte-level BPE, 16,384 tokens = 256 bytes + 16,127 merges + <|bos|>; one token per digit; 4.29 bytes per token
Training data	10,792,862,025 uint16 tokens from 9,670,189 documents

Contents

1 Data pipeline 2

2 Data source 2

3 Held-out validation set 2

4 Tokenizer 3

5 Packed training data 4

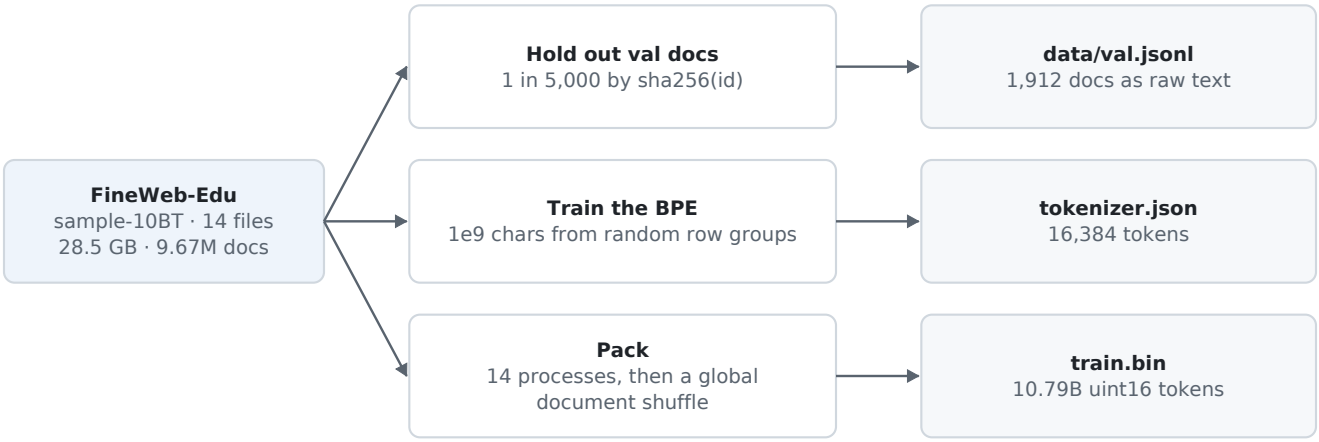
6 Model 5

7 FLOP accounting 5

8 Training 5

9 Evaluation 5

1 Data pipeline



The pipeline runs every step in order: download, hold out validation documents, train the tokenizer, pack. A finished step is skipped on a rerun, so building a second tokenizer reuses the download and the validation set.

2 Data source

The corpus is HuggingFaceFW/fineweb-edu, subset sample-10BT: 14 parquet files, 28.5 GB, 9,672,101 documents and about 10B GPT-2 tokens. Each row group holds 1,000 documents. The code reads two columns, `id` and `text`. `huggingface_hub.snapshot_download` fetches the files; it resumes after an interruption and skips files already present. A single source keeps the data distribution stationary.

> **Finding.** Each file is sorted by Common Crawl dump. In file 000, row group 0 is all CC-MAIN-2013-20, the middle row group is CC-MAIN-2017-30 and the last is CC-MAIN-2023-23. Anything that reads the files in order sees years of web drift. The validation split, the tokenizer's training text and the training order are each built so that file order does not matter.

3 Held-out validation set

- **Selection.** A document is held out when the first 8 bytes of sha256(id), read as a little-endian integer, are divisible by 5,000 (`data.is_val`). The choice depends only on the id, so it is independent of file order and covers every dump.
- **Size.** 1,912 documents and 8,569,151 UTF-8 bytes, about 2.0M tokens with the 16K tokenizer.
- **Format.** Raw text in `data/val.jsonl` as `{"id", "text"}` lines. Any tokenizer can re-tokenize the same bytes.
- **Exclusion.** The tokenizer's training text and train.bin both filter with the same `is_val`, so no held-out document is ever trained on.
- **Cost.** Writing the set reads only the id column of each row group. Text is decoded only for the row groups that contain a held-out id, so the whole pass takes about 25 s.

4 Tokenizer

4.1 The BPE algorithm

Byte-pair encoding builds a vocabulary bottom-up from the 256 byte values. The implementation is the Rust BpeTrainer in HuggingFace tokenizers 0.22.2, wrapped by `mini_gpt/tokenizer.py`.

Training

1. Split the text into pre-tokens with the regex in 4.2. Map each byte to a printable character with the GPT-2 byte table, for example space to Ġ and newline to Ċ.
2. Count how often each distinct pre-token occurs. Start the vocabulary with the 256 byte symbols.
3. Find the adjacent symbol pair that occurs most often across all pre-tokens, weighted by pre-token frequency. Add the merged symbol to the vocabulary, record the merge, and replace the pair everywhere. Repeat.
4. Stop at 16,383 symbols: 256 bytes plus 16,127 merges. A token's ID is its order of creation, so ID 256 is the first merge.

Encoding and decoding

Encoding pre-tokenizes the same way. Within each pre-token it starts from single bytes and applies the learned merges in priority order, earliest-learned first, until none applies. Merges never cross a pre-token boundary, and that is what the regex controls. Decoding concatenates each token's bytes. Every byte value is in the base alphabet, so any UTF-8 string encodes with no unknown token and decodes back to exactly the same string.

Learned on our corpus	Tokens
First six merges	Ġ+t Ġ+a h+e i+n r+e Ġt+he (“ the”, ID 261)
Last merges	Ġpersu+asive Ġnic+otine Ġneg+lected Ġcha+os
Longest tokens	“ responsibilities”, “ gastrointestinal”, “ characteristics”, “ recommendations”

4.2 Pre-tokenization

```
(?i:'s|'t|'re|'ve|'m|'ll|'d)     contractions, case-insensitive
|[^\r\n\p{L}\p{N}]?\p{L}+        a letter run with one optional leading char: " the"
|\p{N}                          ONE digit (GPT-4 / Llama 3 use \p{N}{1,3})
| ?[^\s\p{L}\p{N}]+[\r\n]*      punctuation, optional leading space, trailing newlines
|\s*[\r\n]+                      newline runs
|\s+(?!\S)|\s+                   other whitespace; the last space joins the next word
```

The pattern is the GPT-4 / Llama 3 split, with `\p{N}{1,3}` narrowed to `\p{N}`. A digit is then always a pre-token of its own, so no merge can join two digits. After the split, `ByteLevel(add_prefix_space=False, use_regex=False)` only maps bytes to characters. It adds no leading space and runs no second regex, so encode then decode is the identity. The GPT-4 pattern is preferred over GPT-2's because it matches contractions in any case, attaches one leading symbol to a word, and groups newlines.

Input	Tokens, separated by spaces (␣ is a literal space; <XX> is a raw byte in hex)
`In 1969, 3.5% of <|bos|> users`	In ␣ 1 9 6 9 , ␣ 3 . 5 % ␣of ␣< | b os | > ␣users
`Photosynthesis converts 6 CO2`	Ph ot osynthesis ␣conver ts ␣ 6 ␣CO 2
`naïve café`	na <C3> <AF> ve ␣ca f é — “ï” is rare in the corpus and stays as two byte tokens

Input	Tokens, separated by spaces (␣ is a literal space; <XX> is a raw byte in hex)
two CJK characters (6 bytes)	<E4> <B8> <AD> <E6> <96> <87> — CJK is rare in FineWeb-Edu and stays close to raw bytes

4.3 Vocabulary size

The vocabulary has 16,384 = 2^{14} tokens, including `<|bos|>`. Every ID fits in uint16, which halves disk and page-cache use against int32. At small model widths the embedding matrix is a large share of the parameters, so the vocabulary is kept small. On the validation set the tokenizer averages **4.29 bytes per token**, against 4.63 for GPT-2's 50,257-token vocabulary: 8% more tokens from a third of the vocabulary, partly because digits are split.

4.4 The document marker

`<|bos|>` is the only special token. Its ID is 16,383, one past the BPE vocabulary, and it is never registered with the HuggingFace backend. HuggingFace matches registered special tokens inside raw text even with `add_special_tokens=False`, so a document containing the literal string `<|bos|>` would otherwise produce a control token. Outside the backend, encoding cannot produce it: packing inserts it before each document, and decoding drops it.

4.5 Training text

The tokenizer trains on the first 10^9 characters of training documents. The documents come from row groups visited in a seeded random order (seed 0), because reading files from the start would see only 2013 crawls. Held-out documents are skipped. Training takes about 50 s on 16 cores. `Tokenizer.train` raises an error if the corpus is too small to reach the requested vocabulary size.

4.6 Identity

`Tokenizer.fingerprint()` is the sha256 of the serialized tokenizer. Packing writes it into `meta.json`, so packed data can always be matched to the tokenizer that produced it.

5 Packed training data

5.1 Format

`data/v16k/train.bin` is one flat uint16 array: for each document, `<|bos|>` followed by its tokens. There is no padding, no truncation and no other separator. It holds 10,792,862,025 tokens (21.6 GB) from 9,670,189 documents, an average of 1,116 tokens per document. `meta.json` records the source, the tokenizer's sha256, the vocabulary size, the `<|bos|>` ID, the document and token counts, and the shuffle seed. The marker goes before each document, not after, so every document's first token is predicted from the same symbol.

5.2 Global document shuffle

The training documents are written in one random permutation (seed 0). Without the shuffle, reading the files in order would show 2013 crawls first and 2024 crawls last, so the data distribution would drift during training. 14 processes each tokenize one file into a temporary part, then the documents are gathered from the parts in permuted order. The output goes to `train.bin.tmp` and is renamed only when complete, so an interrupted pack never leaves a file that looks finished.

5.3 Time and disk

Packing sample-10BT took 6.2 hours on 16 cores. The 28.5 GB of parquet files are kept so they can be re-tokenized. train.bin needs 21.6 GB per tokenizer, and the temporary parts need the same again during packing.

6 Model

7 FLOP accounting

8 Training

9 Evaluation